

BiSTNet: Semantic Image Prior Guided Bidirectional Temporal Feature Fusion for Deep Exemplar-Based Video Colorization

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Abstract—How to effectively explore the colors of exemplars and propagate them to colorize each frame is vital for exemplar-based video colorization. In this article, we present a BiSTNet to explore colors of exemplars and utilize them to help video colorization by a bidirectional temporal feature fusion with the guidance of semantic image prior. We first establish the semantic correspondence between each frame and the exemplars in deep feature space to explore color information from exemplars. Then, we develop a simple yet effective bidirectional temporal feature fusion module to propagate the colors of exemplars into each frame and avoid inaccurate alignment. We note that there usually exist color-bleeding artifacts around the boundaries of important objects in videos. To overcome this problem, we develop a mixed expert block to extract semantic information for modeling the object boundaries of frames so that the semantic image prior can better guide the colorization process. In addition, we develop a multi-scale refinement block to progressively colorize frames in a coarse-to-fine manner. Extensive experimental results demonstrate that the proposed BiSTNet performs favorably against state-of-the-art methods on the benchmark datasets and real-world scenes. Moreover, the BiSTNet obtains one champion in NTIRE 2023 video colorization challenge (Kang et al. 2023).

Index Terms—Exemplar-based video colorization, deep convolutional neural network, semantic image prior, bidirectional fusion.

I. INTRODUCTION

VIDEO colorization involves generating fully colorized videos from their grayscale (monochrome) versions. Due to the low quality of film technology in past decades, many black-and-white movies were captured, making this problem

a recent area of attention. As an ill-posed problem with only monochrome videos provided, video colorization is more challenging than single-image colorization as it requires high-quality frames and better temporal consistency.

One of the simplest approaches for video colorization is to employ an image-based colorization model to process each frame independently. In recent years, numerous single-image colorization methods have been proposed, which have achieved remarkable advancements [2], [7], [8], [9], [10], [11], [12]. However, these methods are often associated with temporal flickering artifacts and discontinuities. Consequently, the need for high-quality colorized videos that appear natural and are devoid of flickering artifacts has posed a significant challenge for video colorization.

Several automatic colorization methods have been proposed to constrain temporal consistency in video colorization tasks. These include methods that utilize self-regularization and temporal constraints [3], conditional generative adversarial networks (GANs) [13], and bidirectional deep feature propagation [6]. However, these methods primarily focus on constraining temporal consistency, and the colorization performance of each frame is often unsatisfactory. Furthermore, relying solely on training data is insufficient for colorization because it is an ill-posed problem. As a result, exemplar-based methods have recently garnered more attention as another category of approaches.

Exemplar-based methods for video colorization, such as [4], [5], [14], [15], utilize a reference image to transfer colors to grayscale frames. In contrast to automatic methods, exemplar-based approaches leverage the color information of a reference exemplar to colorize every frame within the entire colorization sequence, enabling flexibility in generating any desired style. However, these methods tend to struggle with long or complex real-world scenes, where the objects present in a single reference image may not cover all those in the video. As a result, wrong colors may be transferred, leading to unsatisfactory visual performance. While increasing the number of high-quality exemplars can alleviate this issue, reducing the number of exemplars is preferred in real-world scenarios where ground truth exemplars are usually unavailable. To the best of our knowledge, only a few works have explored the number of exemplars in real-world video colorization. Therefore, there is a need for an effective video colorization network that is particularly designed for long

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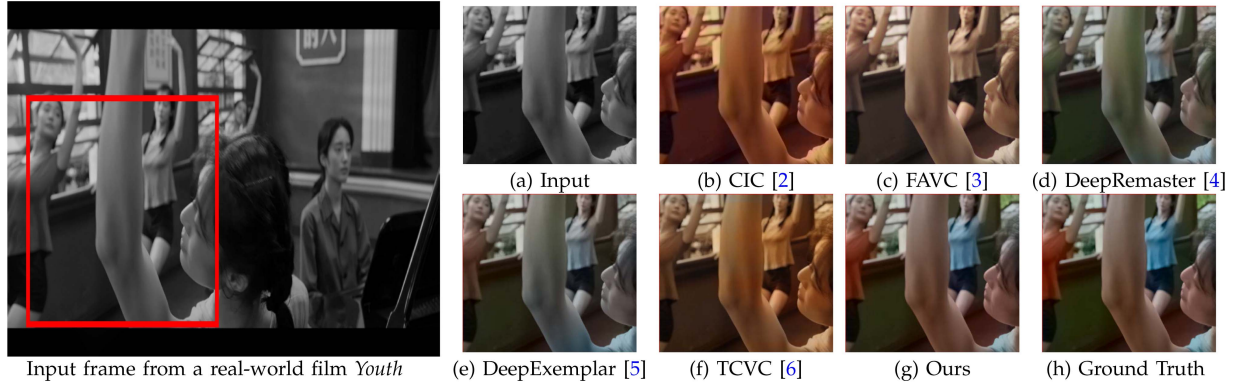


Fig. 1. Colorization results on a challenging film *Youth*. Our method explores well-colored pixels from reference exemplars by a semantic correspondence and adaptively propagates the estimated colors by a bidirectional temporal feature fusion under the guidance of the semantic image prior to achieve better video colorization. As our analysis shows, reducing the influence of the inaccurately matched colors with the proposed bidirectional temporal feature fusion module and utilizing semantic information to guide the colorization process can obtain better colorized results than both the exemplar-based methods [4], [5] and the automatic colorization methods [2], [3], [6] in terms of accuracy and temporal consistency.

videos, maintain colorization performance for individual frames, and enhance temporal consistency.

In this article, we propose an effective exemplar-based deep video colorization method that leverages semantic correspondence between each frame and reference exemplars in a deep feature space to better explore colors and propagate them for frame colorization. Unlike existing methods that directly use matched colors to colorize each frame, we develop a bidirectional temporal feature fusion that mitigates the impact of inaccurate estimated colors and enhances color propagation from exemplars. Furthermore, to address the color-bleeding artifact present in most existing methods, we incorporate semantic information to localize object boundaries and guide colorization for these regions. The proposed method is formulated into a multi-scale refinement framework that colorizes frames in a coarse-to-fine manner. Additionally, we discuss the necessity of two exemplars in Section V. Both quantitative and qualitative evaluations demonstrate the effectiveness of our method, which outperforms state-of-the-art methods (see Fig. 1).

The main contributions are summarized below:

- We present an effective exemplar-based video colorization method that leverages both spatial and temporal information for better colorization results.
- We propose a bidirectional temporal fusion block (BTFFB) that integrates bidirectional semantic correspondence based on temporal clues. By mitigating the influence of inaccurate colors from reference exemplars and exploiting temporal information, the BTFFB significantly enhances the overall quality of colorization results.
- To overcome the color bleeding and spilling that often arise near object boundaries during video colorization, we propose a mixed expert block (MEB) that utilizes both semantic and edge information as prior knowledge to guide the colorization process.
- We introduce a multi-scale refinement block (MSRB) that progressively colorizes frames in a coarse-to-fine manner.
- We evaluate the proposed method on synthetic datasets and real-world videos and show that it achieves favorable performance in terms of colorization accuracy and temporal consistency compared to state-of-the-art methods.

Our approach is effective and practical for video colorization. It obtains one champion in NTIRE 2023 video colorization challenge [1].

II. RELATED WORK

Interactive Colorization: In the early stage of colorization, local user hints are the most popular [7], [16], [17], [18], [19]. These methods require input from the user either in the form of points, strikes, or scribbles and rely on the assumption that nearby pixels should have similar colors. Levin et al. [7] propose an interactive colorization technique that propagates colors from scribbles to neighboring similar pixels. Sangkloy et al. [20] use an end-to-end feed-forward deep generative adversarial architecture to colorize images. To guide structural information and color patterns, user input in the form of sketches and color strokes is employed. Zhang et al. [11] develop user interaction based on two variants, local hint and global hint networks, both of which utilize a common main branch for image colorization. The non-trivial human effort and aesthetic skills to generate colorful images required by user-guided methods make them unsuitable for video colorizing tasks.

Exemplar-Based Colorization: Exemplar-based colorization controls the color styles of the output frames by using reference frames. [5], [21] aims to provide diverse colors to the same image. The system is composed of two subnetworks: the similarity sub-network and the colorization sub-network. Inspired by stylization characteristics in feature extracting and blending, Xu et al. [14] propose a stylization-based architecture to reduce high time and resource consumption. In those works, the correspondence and the color propagation are optimized independently, therefore visual artifacts tend to arise due to correspondence errors.

Video Colorization: Video colorization methods [3], [5], [14], [22], [23] need to consider both colorization performance and temporal consistency. Zhang et al. [5] unify the semantic correspondence and colorization into a single network and train it end-to-end to produce temporal consistent video colorization with realistic effects. Iizuka et al. [4] propose DeepRemaster, a fully 3D convolutional method, utilizing source-reference attention

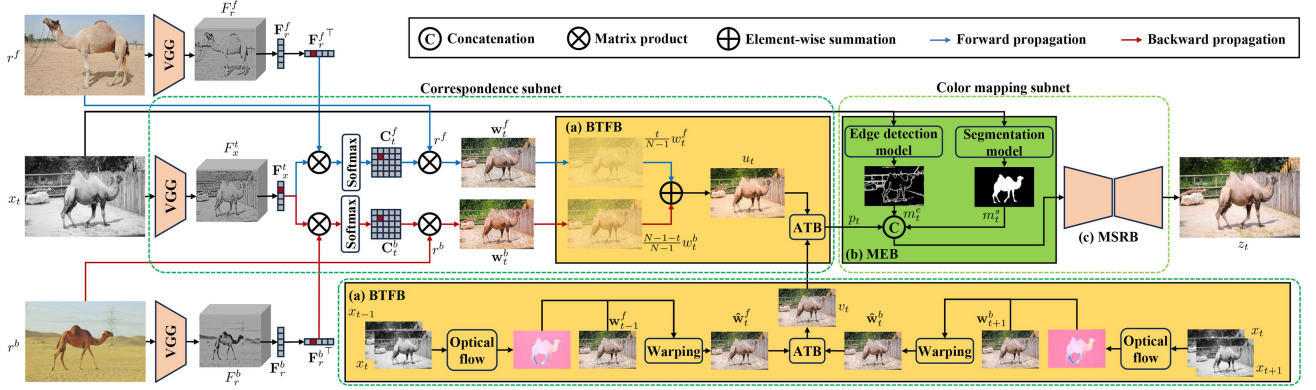


Fig. 2. Architecture of BiSTNet for exemplar-based video colorization. The core components of our method include: (a) bidirectional temporal fusion block (BTFB), (b) mixed expert block (MEB) and (c) multi-scale refinement block (MSRB). We use the pre-trained RAFT model [25] to estimate the optical flow between two consecutive frames. ATB denotes the adaptive temporal blending block proposed by [26], which is employed to blend the temporal features adaptively.

to colorize long videos while maintaining temporal consistency. Recently, FAVC [3] is proposed for automatic video colorization. This method regularizes its model with a KNN graph built on the ground-truth color video and simultaneously posed a temporal loss term for constraining temporal consistency. Lai et al. [24] propose an approach to enforce stronger temporal consistency of a video generated by image processing methods. Different from these methods, we utilize semantic image prior and bidirectional temporal fusion to enhance video colorization.

In this work, we present an effective BiSTNet to explore colors of reference exemplars and utilize them to help video colorization by using a bidirectional temporal feature fusion with the guidance of semantic image prior.

III. PROPOSED APPROACH

Our main objective is to formulate an efficient methodology that explores both spatial and temporal information in order to produce vivid videos with superior temporal consistency. To achieve this, we introduce several essential components, including a bidirectional temporal fusion block (BTFB) that effectively exploits temporal information, a mixed expert block (MEB) that utilizes semantic image prior to guide colorization of regions with object boundaries, and a multi-scale refinement block (MSRB) that performs color mapping in a coarse-to-fine mechanism, resulting in better colorization performance compared to single-scale networks [5], [14]. In this section, we present the overall pipeline of our video colorization network architecture, followed by a detailed description of each component, and conclude with an overview of the loss functions.

Overall Pipeline: Given a grayscale video $X = \{x_0, x_1, \dots, x_{N-1}\}$, where $x_t \in \mathbb{R}^{H \times W \times 1}$ denotes the frame at time t , $H \times W$ denotes the spatial resolution, our main goal is to estimate the colorized video frames $Z = \{z_0, z_1, \dots, z_{N-1}\}$, where $z_t \in \mathbb{R}^{H \times W \times 3}$. We use two reference images $r^f \in \mathbb{R}^{H \times W \times 3}$ and $r^b \in \mathbb{R}^{H \times W \times 3}$ for each video clip. r^f and r^b contain similar contents to the input frames at time 0 and time $N - 1$, respectively.

Specifically, we first apply the correspondence subnet that contains a forward stage and a backward stage to calculate bidirectional semantic correspondence given input grayscale image x_t (see Fig. 3 for the detailed network architectures). The forward semantic correspondence is to obtain the relationship between x_t and r^f and the backward semantic correspondence is to obtain the relationship between x_t and r^b . We establish these semantic correspondence using the deep features F_x^t , F_r^f and $F_r^b \in \mathbb{R}^{H \times W \times C}$ that are extracted from x_t , r^f and r^b by the pre-trained VGG19 model [29]. Then we reshape these features into feature vectors $\mathbf{F}_x^t$, $\mathbf{F}_r^f$ and $\mathbf{F}_r^b \in \mathbb{R}^{HW \times C}$. We estimate the correspondence matrix by

$$\begin{aligned} C_t^f &= \text{softmax} \left(\frac{\mathbf{F}_x^t \mathbf{F}_r^{f\top}}{\|\mathbf{F}_x^t\|_2 \|\mathbf{F}_r^f\|_2} \right), \quad C_t^f \in \mathbb{R}^{HW \times HW} \\ C_t^b &= \text{softmax} \left(\frac{\mathbf{F}_x^t \mathbf{F}_r^{b\top}}{\|\mathbf{F}_x^t\|_2 \|\mathbf{F}_r^b\|_2} \right), \quad C_t^b \in \mathbb{R}^{HW \times HW}, \end{aligned} \quad (1)$$

where $\text{softmax}(\cdot)$ denotes the softmax operation that is applied to each row of the matrix. We then generate the warped results of reference images r^f and r^b by

$$\begin{aligned} \mathbf{w}_t^f &= C_t^f \mathbf{r}^f, \\ \mathbf{w}_t^b &= C_t^b \mathbf{r}^b, \end{aligned} \quad (2)$$

where $\mathbf{r}^f$ and $\mathbf{r}^b$ denote the vector forms of the reference images r^f and r^b . By applying a reshaping function to $\mathbf{w}_t^f$ and $\mathbf{w}_t^b$, we can get the warped color frames $w_t^f \in \mathbb{R}^{H \times W \times 3}$ and $w_t^b \in \mathbb{R}^{H \times W \times 3}$. Taking these two warped frames as input, the BTFB is developed to reduce the impact of inaccurate colors in w_t^f and w_t^b and alleviate color contamination resulting from large motion or severe occlusion (see details in Section III-A).

Color-bleeding artifacts in the vicinity of vital objects in videos represent a prevalent issue in the field of video colorization. We aim to improve the quality of colorized videos by incorporating a color mapping subnet within our BiSTNet. This subnet is comprised of MEB and MSRB which are explained in Sections III-B and III-C, respectively. The subnet generates a

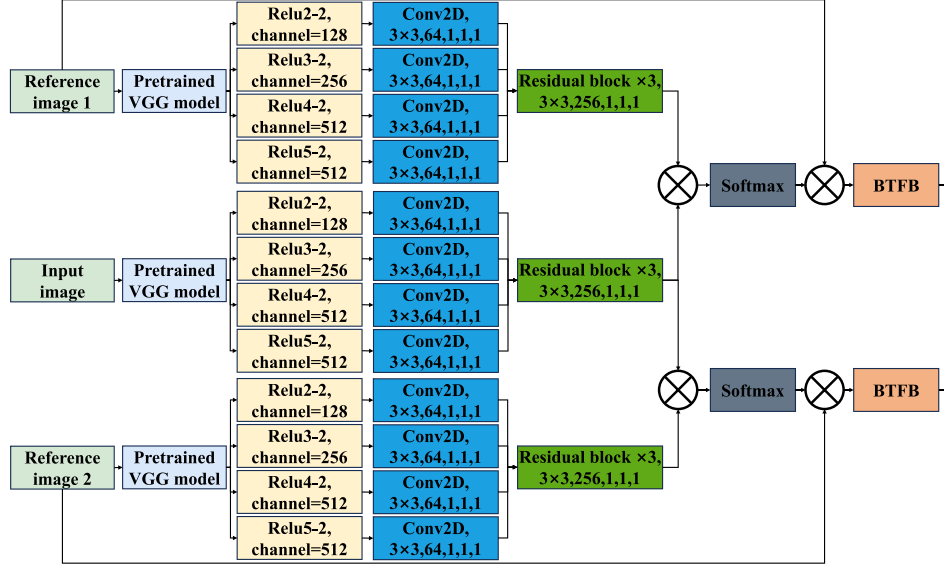


Fig. 3. Network details of the correspondence subnet. The numbers in each convolutional layer and residual block denote the filter size, filter number, stride value, padding value and dilation value.

well-colored image z_t based on two inputs: the grayscale input x_t and the fused color map p_t from the bidirectional temporal fusion block. Given x_t , we employ MEB, which comprises of a semantic segmentation module that outputs a segmentation mask m_t^s and an edge detection module that outputs an edge detection mask m_t^e . We further introduce MSRB to boost performance visually. MSRB is a three-level Unet-based network that uses the grayscale input x_t , fused color map p_t , segmentation mask m_t^s , and edge detection mask m_t^e as inputs to create a well-colored image z_t . A schematic of the BiSTNet pipeline is demonstrated in Fig. 2.

A. Bidirectional Temporal Fusion Block

The quality of colorized video frames is prone to rapid degradation in exemplar-based video colorization methods when future frames fail to match the semantic contents of reference frames. This scenario is common, and to mitigate the resultant mismatch of semantic correspondence, it becomes essential to use temporal clues to identify regions suitable for colorization.

To this end, we introduce a bidirectional temporal fusion block (BTFB) to acquire the fused color map u_t for time t using the estimated warped color images w_t^f and w_t^b obtained as described previously. The fusion of w_t^f and w_t^b is mainly based on the temporal distance between the input frame x_t and the reference images r_f and r_b . We designate the temporal distance from input frame $x_t \in \{x_0, x_1, \dots, x_{N-1}\}$ to forward reference image r^f as $d_t^f = t$ and from input frame x_t to backward reference image r^b as $d_t^b = N - 1 - t$. We normalize these temporal distances based on the maximum distance $N - 1$ between input frames to form weight parameters, denoted as $\tilde{d}_t^f = \frac{t}{N-1}$ and $\tilde{d}_t^b = \frac{N-1-t}{N-1}$. Utilizing $\tilde{d}_t^f$ and $\tilde{d}_t^b$ allows for the acquisition of more accurate fused color maps u_t as corresponding weights for $w_t^f(w_t^b)$ will be larger when x_t is closer to $r_f(r_b)$. The fused

color map u_t can be determined by

$$u_t = \frac{t}{N-1} w_t^f + \frac{N-1-t}{N-1} w_t^b. \quad (3)$$

Note that BTFB differs from [30] in weights assignment and the emphasis on temporal clues over multi-channel fusion.

To further explore temporal clues, we employ the adaptive temporal blending (ATB) block proposed by [26]. Our approach involves estimating the optical flow between two consecutive frames using the RAFT method [25], and then using this information to generate the color frames $\hat{w}_t^f(\hat{w}_t^b)$ from $w_{t-1}^f(w_{t+1}^b)$. We then adopt ATB to adaptively blend the original features $w_t^f(w_t^b)$ with the features $\hat{w}_t^f(\hat{w}_t^b)$ propagated from the previous frames, resulting in the fused feature maps $\bar{w}_t^f(\bar{w}_t^b)$. Note that $\bar{w}_t^f$ benefits from the information provided by frame 0 to frame t in the forward stage. Similarly, $\bar{w}_t^b$ benefits from the information provided by frame $N - 1$ to frame t in the backward stage. Finally, we obtain the output of our bidirectional temporal fusion block p_t by concatenating u_t and v_t , where v_t is obtained by utilizing ATB block to fuse $\bar{w}_t^f$ and $\bar{w}_t^b$.

B. Mixed Expert Block

Existing video colorization methods [3], [4], [5], [6] often suffer from the problem of color-bleeding artifacts, where colors spread improperly across the boundaries of adjacent objects, resulting in poor-quality videos. This limitation hinders the practical use of such methods. To overcome this issue, it is essential to incorporate prior knowledge to improve the modeling of object boundaries and guide the colorization process more effectively.

In this article, we develop a mixed expert block (MEB) to achieve more precise modeling of the boundaries that separate adjacent objects. Our MEB consists of two models: a semantic

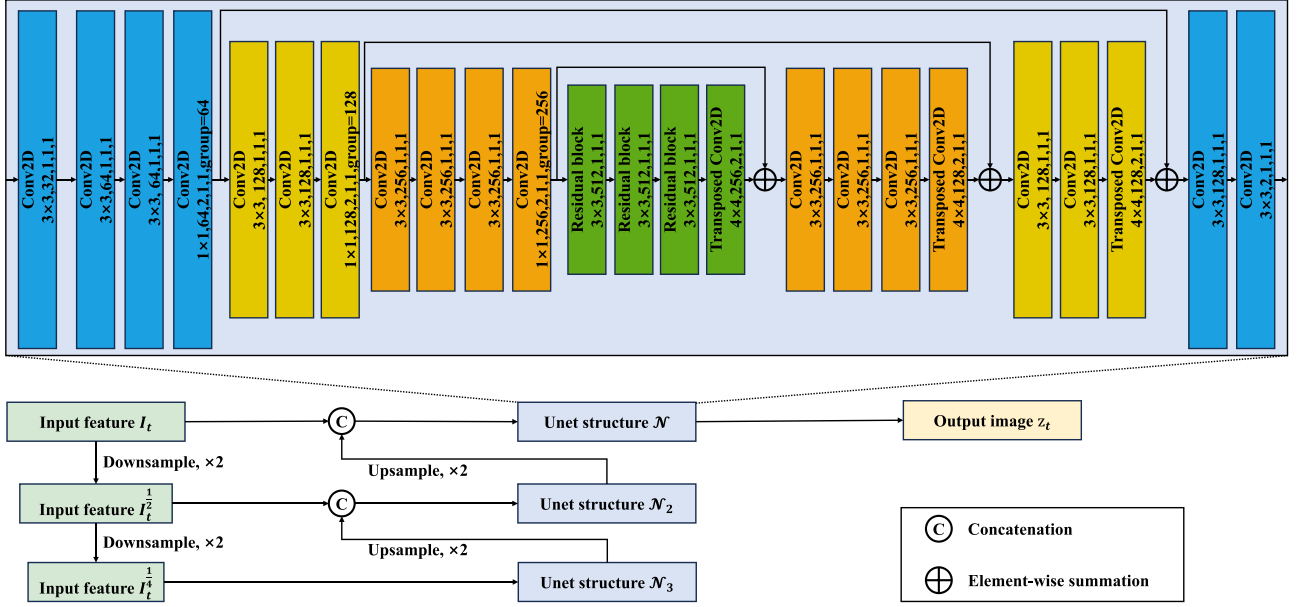


Fig. 4. Network details of the multi-scale refinement block. The numbers in each convolutional layer, residual block, and transposed convolutional layer denote the filter size, filter number, stride value, padding value, dilation value and depth-wise convolution. We use PReLU [27] as activation function within the residual blocks and use instance normalization [28] after each convolutional layer.

segmentation model and an edge detection model. The MEB takes the grayscale frame x_t as input and outputs two masks, a probability mask for semantic segmentation, denoted as m_t^s in Fig. 2, and an edge detection mask, denoted as m_t^e . Specifically, we employ an off-line pre-trained semantic segmentation model [31] to generates $m_t^s \in \mathbb{R}^{H \times W \times C_{Seg}}$, where C_{Seg} represents the number of semantic labels. Additionally, an off-line pre-trained edge detection model [32] is utilized to produce $m_t^e \in \mathbb{R}^{H \times W \times 1}$. Finally, the output of MEB, m_t^s and m_t^e serve as vivid semantic prior to guide our BiSTNet, resulting in more precise colorization performance.

C. Multi-Scale Refinement Block

To further improve the quality of the colorized videos, we employ a multi-scale refinement block (MSRB) based on the commonly used coarse-to-fine strategy [33], [34]. MSRB contains three levels, each level is built by the same Unet structure (denoted as $\mathcal{N}$, $\mathcal{N}_2$, $\mathcal{N}_3$). It takes the concatenation results of the grayscale input x_t , the fused color map p_t , the segmentation mask m_t^s and the edge detection mask m_t^e as input. Specifically, let I_t denote the input of MSRB, we first downsample I_t to the $1/2$ resolution feature $I_t^{\frac{1}{2}}$ and a $1/4$ resolution feature $I_t^{\frac{1}{4}}$. Second, we apply a $\mathcal{N}_3$ to the feature $I_t^{\frac{1}{4}}$ and obtain a coarse colorized frame $z_t^{\frac{1}{4}}$. In this level, our target is to train $\mathcal{N}_3$ good at controlling the global color style. For the second level, the network $\mathcal{N}_2$ takes the concatenation of $I_t^{\frac{1}{2}}$ and upsampled colorized frame $z_t^{\frac{1}{4}}$ as input and generates the finer colorized frame $z_t^{\frac{1}{2}}$. Similarly, the network $\mathcal{N}$ takes the concatenation of the original resolution feature I_t with the upsampled result of

$z_t^{\frac{1}{2}}$ as input and restores the colorized frame z_t at the original resolution (see Fig. 4 for the detailed network architectures).

D. Loss Function

To better constrain the network training, we use the edge-enhancing loss by [36] to alleviate the color-bleeding artifacts and the problematic color spreading near the boundaries between adjacent objects

$$\mathcal{L}_{edge} = \|\mathcal{S}(y_t) - \mathcal{S}(z_t)\|_2, \quad (4)$$

where $\mathcal{S}$ is the Sobel filter used in [36].

We note that the majority of the colorized areas are smooth, artifacts usually happen at object boundaries. Simply using the L_1 -norm-based content aware loss function (i.e., $\|y_t - z_t\|_1$, where z_t denotes the colorized frame produced by our BiSTNet and y_t denotes the ground truth frame.) would make the trained colorization model mainly focus on well-colorized regions and ignore the regions where artifacts exist. To overcome this problem, we further employ the hard example mining loss by [33], [37].

In addition to those above loss functions, we further use the same loss function based on the content, perception, adversarial and temporal consistency by [5] to constrain the network training. Finally, the loss function for training is

$$\mathcal{L}_{total} = \lambda_{edge}\mathcal{L}_{edge} + \lambda_{hem}\mathcal{L}_{hem} + \lambda_c\mathcal{L}_c, \quad (5)$$

where $\mathcal{L}_c$ is the loss function by [5], $\mathcal{L}_{hem}$ is the hard example mining loss function by [33]. λ_{edge} , λ_{hem} , and λ_c are weight parameters.

TABLE I
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD AGAINST STATE-OF-THE-ART ONES ON THE SYNTHETIC DAVIS DATASET [35]

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CDC $\downarrow$	FID $\downarrow$	Colorful $\uparrow$
lizuka et al. [10]	23.88	0.947	0.176	0.005461	99.13	18.64
CIC [2]	22.57	0.947	0.106	0.006746	113.48	31.11
CIC [2]+DVP [45]	22.44	0.945	0.101	0.005346	102.49	30.19
IDC [11]	24.88	0.949	0.116	0.004833	102.77	18.21
InsColor [12]	25.65	0.951	0.082	0.008267	98.02	20.94
InsColor [12]+DVP [45]	25.53	0.950	0.106	0.004522	92.77	22.11
Gray2ColorNet [44]+DVP [45]	27.24	0.965	0.054	0.004323	97.56	16.88
Color2Embed [43]+DVP [45]	27.65	0.969	0.049	0.004227	89.72	13.23
FAVC [3]	25.98	0.967	0.172	0.004096	96.99	19.57
TCVC [6]	26.34	0.962	0.175	0.003947	113.08	18.96
DeepRemaster [4]	27.03	0.964	0.057	0.004725	86.53	16.94
DeepRemaster*	28.38	0.967	0.046	0.004611	71.31	20.18
DeepExemplar [5]	28.64	0.972	0.041	0.004469	53.47	21.09
DeepExemplar*	29.12	0.974	0.034	0.004334	41.10	21.30
Ours single-reference	29.07	0.973	0.036	0.003846	39.76	23.53
Ours double-reference	30.40	0.976	0.030	0.003540	36.34	26.23

We use the notation * to indicate that the model is the same as before, but with the use of two reference frames. DeepRemaster [4] allows for the use of multiple reference frames. For DeepExemplar*, we modify the color warping layer of the DeepExemplar model [5] by computing a correlation matrix that characterizes the similarity between the deep features of the input frames and two reference frames, instead of just one reference frame.

Our method achieves the best performance in most of the metrics.

The results of the best performance are denoted in bold.

IV. EXPERIMENTAL RESULTS

In this section, we evaluate the proposed method against state-of-the-art ones. Due to the page limit, we include more results in the supplemental material. The training code and test model are available at <https://github.com/yyang181/BiSTNet>.

Datasets: We adopt the DAVIS dataset [35] and the Videvo dataset [24] as the benchmark datasets for training and testing. Our training set includes 60 clips from DAVIS and 80 clips from Videvo, whereas our testing set includes 30 clips from DAVIS and 20 clips from Videvo. Before training, we resize all frames of the training videos to a dimension of 384×224 pixels. For the purpose of fair comparison, we maintain the original resolution of the test videos when evaluating the performance of our method and other methods.

To obtain the reference color images, we follow a similar strategy as in [4]. Specifically, we use the first and the last frames of each video clip as the reference images. We then employ these reference images to colorize the remaining frames. To prevent introducing ground truth information during testing, we exclude the first 5 frames and the last 5 frames from each video clip as consecutive input frames for evaluation. In all double-reference methods, we choose the first frame and the last frame of the entire video clip as the reference frames. In all single-reference methods, we only utilize the first frame as the reference. For our “single-reference” version, we duplicate the first reference frame to obtain the second reference frame. This allows our “single-reference” version to be trained and tested using the same configuration as our “double-reference” version.

Evaluation Metrics: To evaluate the quality of the colorized videos, we use the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM) [38], the learned perceptual image patch similarity (LPIPS) [39] matrices and the *Fréchet Inception Distance* (FID) [40]. The color distribution consistency

index (CDC) [6] is used to measure the temporal consistency of videos. In addition, we measure the colorfulness using the psychophysics metric from [41].

Implementation Details: We train all models using the PyTorch framework on a machine with four RTX-A6000 GPUs. We adopt the CIE *LAB* color space for each frame in our experiments. In the training process, we use the Adam algorithm [42] with default parameters and train our models with 50 epochs. The batch size is set to 8. The learning rate is set to a constant 2×10^{-4} . We employ the off-the-shelf method ProtoSeg [31] for semantic segmentation. We also employ HED [32] for edge detection with default parameters. For the weights in the loss function, we empirically set $\lambda_{edge} = 2$, $\lambda_{hem} = 2$, and $\lambda_c = 1$.

To ensure fair comparisons, we retrain deep learning-based methods [2], [5], [11], [12], [43] using the official codes on identical datasets as our method. In addition, for [3], [6], whose weights are trained on the same datasets as our method, we conduct tests using their official codes and weights provided by the authors. Moreover, we use their official codes and weights provided by the authors for [4], [10], [44].

A. Comparisons With State-of-the-Art Methods

We compare the proposed method with state-of-the-art ones including both the automatic colorization ones [3], [6] and exemplar-based ones [4], [5]. In addition, to make the article more self-contained, we include comparisons with state-of-the-art image colorization methods [2], [10], [11], [12]. To ensure fair comparisons, we compare ours double-reference model with DeepExemplar* and DeepRemaster*, which are the same models as before but with the use of two reference frames. We also compare ours single-reference variant with DeepExemplar [5] and DeepRemaster [4]. We use DVP [45] as a post-processing step for the results of image colorization methods. We then compare

TABLE II
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD AGAINST STATE-OF-THE-ART ONES ON THE SYNTHETIC VIDEO DATASET [24]

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CDC $\downarrow$	FID $\downarrow$	Colorful $\uparrow$
Lizuka et al. [10]	25.28	0.938	0.113	0.005283	54.15	22.23
CIC [2]	23.15	0.901	0.136	0.005521	69.10	29.03
CIC [2]+DVP [45]	23.10	0.900	0.137	0.004218	67.13	28.66
IDC [11]	24.10	0.933	0.122	0.004517	66.14	20.26
InsColor [12]	25.62	0.945	0.112	0.004007	59.41	24.77
InsColor [12]+DVP [45]	25.23	0.942	0.106	0.003451	55.46	24.17
Gray2ColorNet [44]+DVP [45]	28.31	0.957	0.077	0.002986	50.07	16.38
Color2Embed [43]+DVP [45]	28.40	0.960	0.074	0.002945	49.12	15.51
FAVC [3]	25.47	0.952	0.094	0.010251	54.97	18.18
TCVC [6]	26.37	0.959	0.097	0.001995	53.45	19.77
DeepRemaster [4]	28.81	0.966	0.070	0.002405	43.17	22.43
DeepRemaster*	29.98	0.967	0.066	0.002378	40.31	20.18
DeepExemplar [5]	29.89	0.969	0.060	0.002018	28.30	25.54
DeepExemplar*	30.75	0.970	0.054	0.002011	26.36	25.49
Ours single-reference	31.01	0.973	0.045	0.001957	17.76	26.53
Ours double-reference	32.80	0.977	0.037	0.001916	11.89	26.78

Our method achieves the best performance in most of the metrics.
The results of the best performance are denoted in bold.

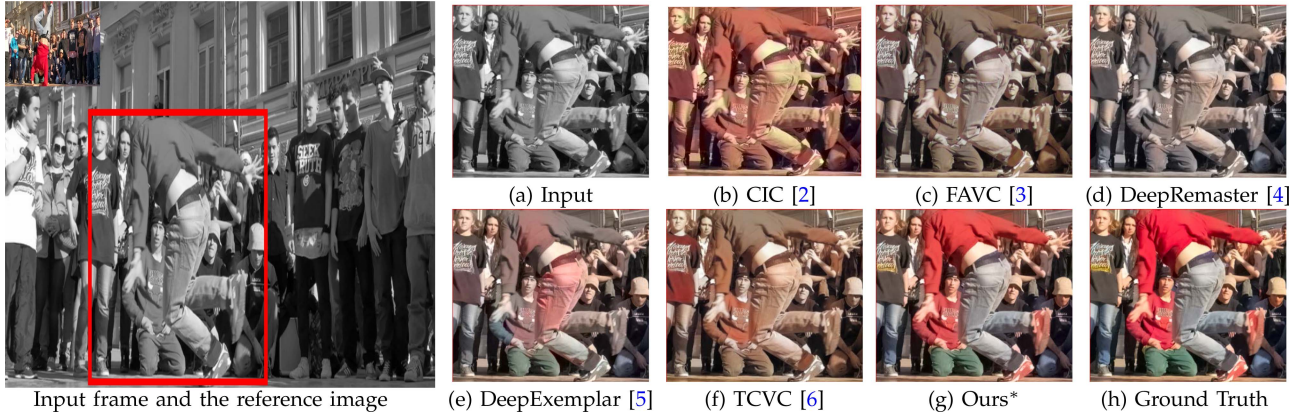


Fig. 5. Qualitative comparisons on the clip *breakdance* from the DAVIS dataset [35]. The evaluated methods do not generate colorful frames, and the colors are not estimated correctly in (b)–(f). Our method generates well-colored image.

the performance of single-reference exemplar-based image colorization methods Gray2ColorNet [44] and Color2Embed [43].

Table I shows the quantitative evaluation results on the commonly used benchmark [35], where the proposed BiSTNet family generates better-colored results. Specifically, ours single reference variant outperforms DeepExemplar [5], a state-of-the-art exemplar-based method, by up to 0.43 dB in terms of PSNR, and achieves 13.94% improvement in terms of the temporal consistency index CDC. Ours double reference outperforms DeepExemplar* by up to 1.28 dB in terms of PSNR, and achieves 18.32% improvement in terms of the CDC. Similarly, Table II shows the quantitative evaluation results on the commonly used benchmark [24], where the proposed BiSTNet family generates better-colored results. All comparisons in Tables I and II demonstrate that our BiSTNet generates better-colored frames while producing videos with better temporal consistency.

Qualitative comparisons of the proposed BiSTNet and top-performing methods listed in Tables I and II are presented in Figs. 5, 6 and 7. It is noteworthy that the DeepExemplar [5] and the DeepRemaster [4] fail to produce accurate color results. In contrast, the proposed BiSTNet generates vivid and visually

accurate frames. Specifically, in Fig. 5, the proposed BiSTNet restores the color of the cloth. Moreover, the proposed BiSTNet effectively prevents color contamination near the boundaries of lawns and roads in Fig. 6, highlighting the efficacy of the proposed MEB in video colorization tasks. In Fig. 7, the proposed BiSTNet restores the color of neon lights and billboards.

B. Evaluations on Real-World Grayscale Videos

We further evaluate the proposed method on real-world grayscale videos, where the ground truth colorful videos are not available. Fig. 9 shows an example from a challenging real-world old film *Bend-Or*. To generate reference exemplars for all exemplar-based methods, DeOldify [46], a non-reference based video colorization method, is used. State-of-the-art methods [2], [3], [4], [5], [6] are compared with the proposed method. Fig. 9 shows that state-of-the-art methods do not colorize the objects (e.g., horse) well. In contrast, our method generates a better-colored frame, where the colors of the horse look natural (see Fig. 9(g)).

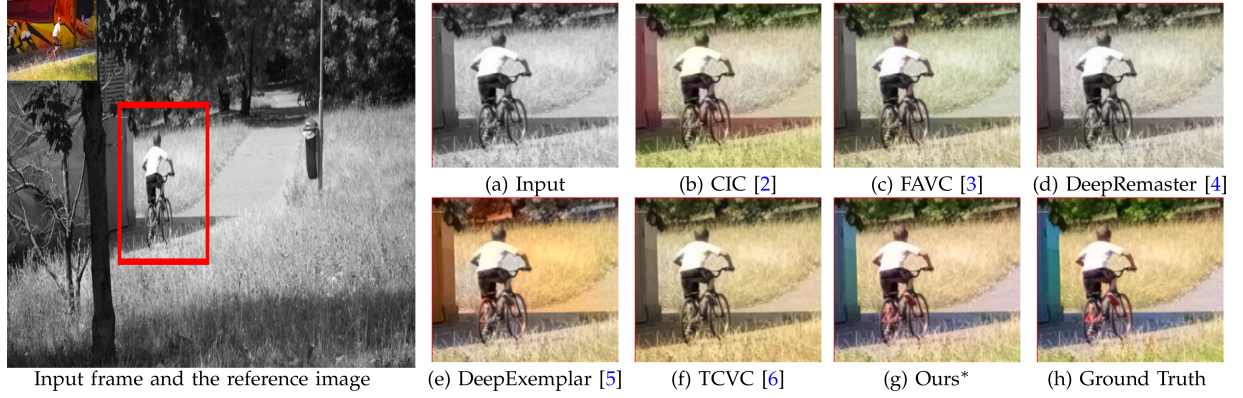


Fig. 6. Qualitative colorization comparisons on the clip *bmx-trees* from the DAVIS dataset [35]. Our method in (g) is able to avoid color contamination near the boundaries of lawns and roads.

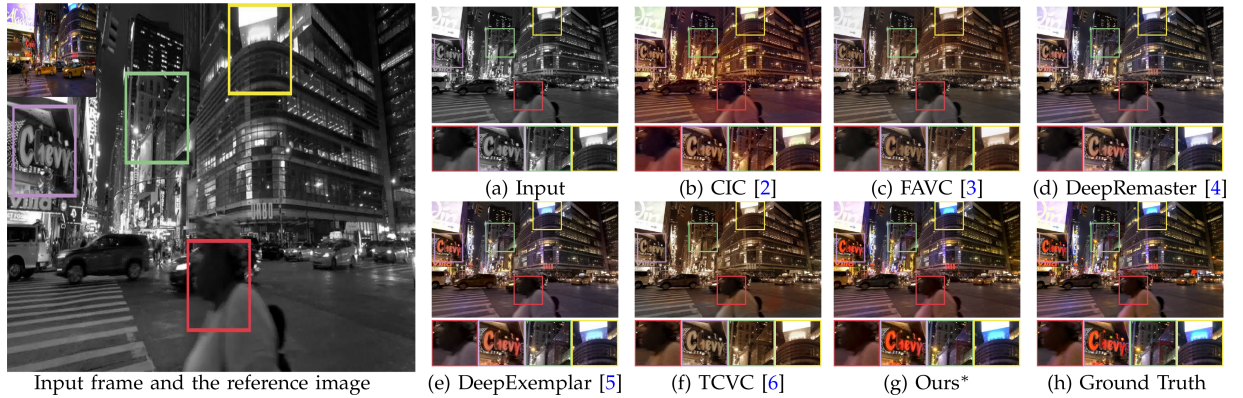


Fig. 7. Qualitative colorization comparisons on the clip *TimeSquareTraffic* from the Videvo dataset [24]. The evaluated methods do not generate colorful frames, and the colors are not estimated correctly in (b)–(f). Our method generates well-colored image.

C. Temporal Consistency Evaluations

Existing methods mainly concentrate on enhancing the quality of individual colorized frames. However, temporal consistency is a critical determinant of video colorization quality. In this study, we investigate the temporal consistency of colorized videos produced by our proposed method. We present a plot of the correlation between the quality of each colorized frame and the temporal consistency in Fig. 8. Our findings indicate that our proposed method generates high-quality colorized videos with superior temporal consistency.

V. ANALYSIS AND DISCUSSION

In this section, we further conduct extensive ablation studies to demonstrate the effectiveness of the proposed temporal fusion model, mixed expert block, and multi-scale refinement block. We train all the possible baseline models using the same settings as the proposed BiSTNet and evaluate them on the DAVIS dataset [35] for fair comparisons. Table III shows the quantitative evaluation results.

Effectiveness of the Bidirectional Temporal Fusion Block: The proposed BTFB is used to utilize temporal clues to select well-colored regions from the reference exemplars. To demonstrate

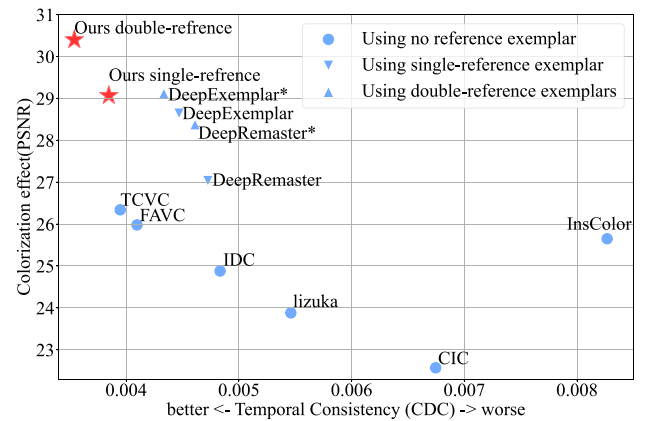


Fig. 8. Model performance and temporal consistency comparison between our proposed BiSTNet family and other colorization methods on the DAVIS dataset [35]. Our BiSTNet family achieves both a higher PSNR and a lower CDC among all other methods.

its effectiveness, we replace the BTFB with average weighting to fuse the bidirectional features. Table III shows that our model without BTFB does not perform well.

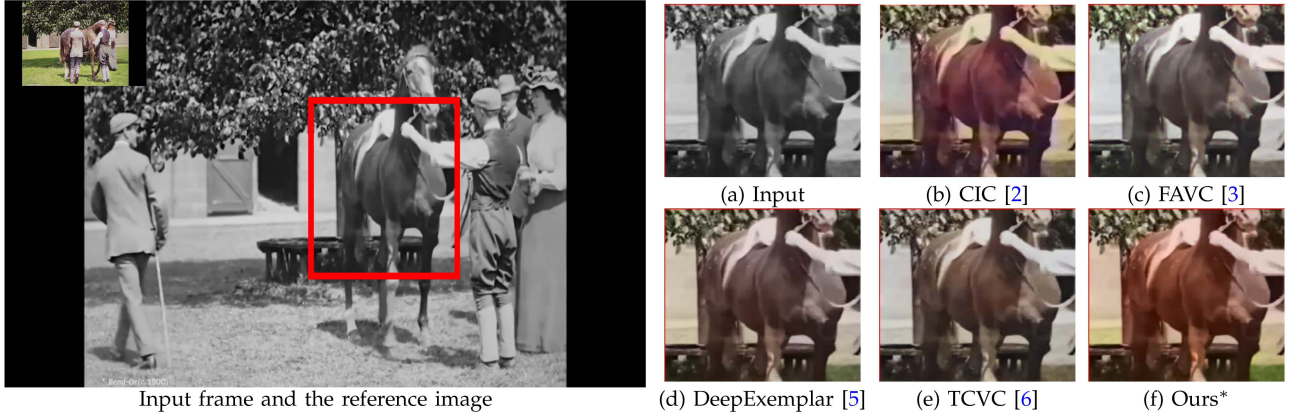


Fig. 9. Qualitative colorization comparisons on the real-world film *Bend-Or*. Methods [3], [5], [6] do not colorize the horse and the backgrounds well. Our method colorizes the frame well, where the colors look better.

TABLE III
QUANTITATIVE EVALUATIONS OF EACH COMPONENT IN THE PROPOSED
BiSTNET ON THE DAVIS DATASET [35]

Method	PSNR $\uparrow$	CDC $\downarrow$
Ours w/o BTFB	29.57	0.004324
Ours w/o m_t^s	29.69	0.003955
Ours w/o m_t^e	29.77	0.003846
Ours w/o m_t^e & m_t^s	29.41	0.003936
Ours w/o MSRB	29.58	0.003620
Ours w/o $\mathcal{L}_{edge}$	29.44	0.003715
Ours w/o $\mathcal{L}_{hem}$	29.38	0.003778
Ours w/o $\mathcal{L}_{edge}$ & $\mathcal{L}_{hem}$	29.40	0.003690
Ours	30.40	0.003540

The results of the best performance are denoted in bold.

Effectiveness of the Mixed Expert Block: The MEB is utilized to exploit image prior, including semantic information and object edges, to guide the colorization process. InsColor [12] employs an off-the-shelf object detector and leverages an instance colorization network to extract object-level features. However, it remains unclear whether incorporating semantic and edge information is advantageous for video colorization. To demonstrate the effectiveness of MEB, we compare it with the baseline model by replacing the segmentation mask m_t^s and the edge mask m_t^e individually. Additionally, we perform an experiment to eliminate both m_t^s and m_t^e . Table III reveals that the colorization models without image prior result in poor video colorization. Fig. 10 visually illustrates the effectiveness of MEB. It is noteworthy that the BiSTNet, without utilizing either m_t^s or m_t^e , produces inadequately colorized frames with significant color-bleeding artifacts, such as spreading T-shirt colors onto the arm. Conversely, by utilizing the proposed MEB, we are able to mitigate this phenomenon and generate superior colorized frames (see Fig. 10(g)).

Effectiveness of the Multi-Scale Refinement Block: The proposed MSRB adopts the coarse-to-fine strategy that has been widely used in image and video restoration [33], [34]. However, it is unclear whether this strategy is effective for video colorization. To validate its effectiveness on video colorization, we compare the proposed method without this block. The

comparisons of “Ours w/o MSRB” and “Ours” in Table III show that using the MSRB is able to improve the performance of video colorization. Specifically, as shown in Fig. 11(c), the colorized frame generated by the proposed method with MSRB has a better global coloring effect compared to the model without MSRB (Fig. 11(b)) where the color of the fence is contaminated by the color of the roof around the right area of the image.

Effectiveness of the Edge Loss and Hard Example Mining Loss: We further examine the effect of the edge loss and the hard example mining on video colorization. The comparisons of the results in Table III (see “Ours w/o $\mathcal{L}_{edge}$ ”, “Ours w/o $\mathcal{L}_{hem}$ ”, “Ours w/o $\mathcal{L}_{edge}$ & $\mathcal{L}_{hem}$ ”, and “Ours”) show that using these two loss functions helps video colorization.

Necessity of Two Reference Exemplars: Exemplar-based colorization methods have been widely used in image and short video colorization, where a single exemplar provides highly content-relevant guidance. However, in real-world video colorization tasks that involve hundreds or thousands of gray frames, using a single exemplar that is dissimilar to many of the input frames becomes inevitable. This results in significant drops in colorization performance for content-irrelevant frames. Therefore, multiple exemplars can greatly improve long video colorization. Employing two key frames at both ends of a video is a popular approach in video restoration. For instance, TCVC [6] uses an anchor frame branch that propagates features extracted from anchor frames to achieve competitive temporal consistency in video colorization. Feature propagation from both ends of a video is also common in video super-resolution methods, such as BasicVSR [49] and BasicVSR++ [50]. Additionally, video interpolation restores realistic intermediate video frames using two consecutive input frames (key frames). Therefore, we use two exemplars inspired by the earlier methods. To further demonstrate the necessity of using two reference frames, we show the visual comparison of single-reference and double-reference version for three methods, DeepExemplar [5], DeepRemaster [4] and our BiSTNet. Our approach utilizing the double-reference version generates the most natural results as shown in Fig. 12.

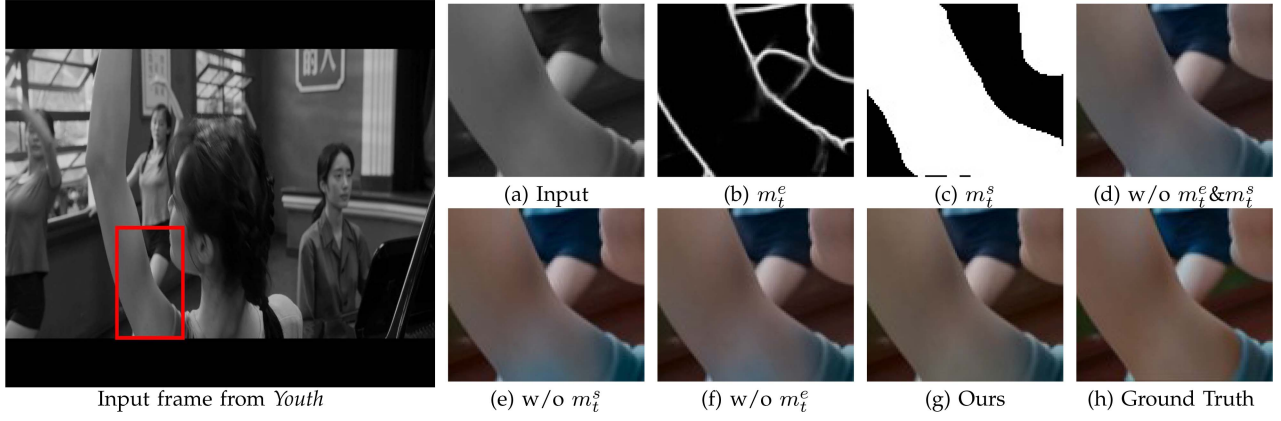


Fig. 10. Effectiveness of the proposed MEB module. (b) and (c) Denote the edge detection mask m_t^e and the semantic segmentation mask m_t^s . (d) Denotes the colorized image by the proposed model without using both m_t^e and m_t^s (i.e., the proposed method without using the MEB module). (e) Denotes the colorized image by the proposed model without using m_t^s . (f) Denotes the colorized image by the proposed model without using m_t^e . Note that the method without using the MEB module does not colorize the frame well, where the color of the arm is contaminated by the color of the T-shirt. Our model with the MEB module restores better colors in (g).



Fig. 11. Effectiveness of the proposed MEB. The result by the method without using the MSRB in (b) contains uneven color distribution from left to right. While the method using the MSRB restores the better-colored frame in (c).



Fig. 12. Qualitative comparisons on the *breakdance* from the DAVIS dataset. We use the notation * to indicate that the model is the same as before, but with the use of two reference frames. The reference images are attached to the top left corner of the input image. The evaluated methods do not generate colorful frames, and the colors are not estimated correctly in (b), (c), (e) and (f). Our method generates well-colored image in (d) and (g).

Exemplar Generation With Image Colorization Methods: With the recent advancements in image colorization methods, such as Deoldify [46], DISCO [48] and DDColor [47], obtaining high-quality exemplars has become easier, making our approach more practical for real-world scenes. The first column in Fig. 13 shows the different exemplars generated by different methods or obtained from “Internet” according to Google Image Search (<https://images.google.com/>). We note that the proposed method

is able to transfer the colors of exemplars to the videos and generates natural and vivid colorization results as shown in the second and fifth rows of Fig. 13.

Sensitivity Analysis on the Exemplars: As our method requires exemplars, one may wonder whether our method is sensitive to the contents (e.g., structures) of the exemplars. To answer this question, we use the color images with different structures as the exemplars to evaluate our method. Fig. 14 shows that although



Fig. 13. Qualitative comparisons on the clip 004 from the test set of NTIER 2023 video colorization challenge [1] with different exemplars by different sources. We evaluate the performance of our BiSTNet with different exemplars generated by three state-of-the-art image colorization methods (Deoldify [46], DDColor [47], DISCO [48]) and obtained from “Internet” (i.e., the color images in the first column). For the exemplars from the colorization methods, we use image colorization methods to colorize the first frame of the videos as exemplars. The generated colorization results are shown in the second to the fifth columns.

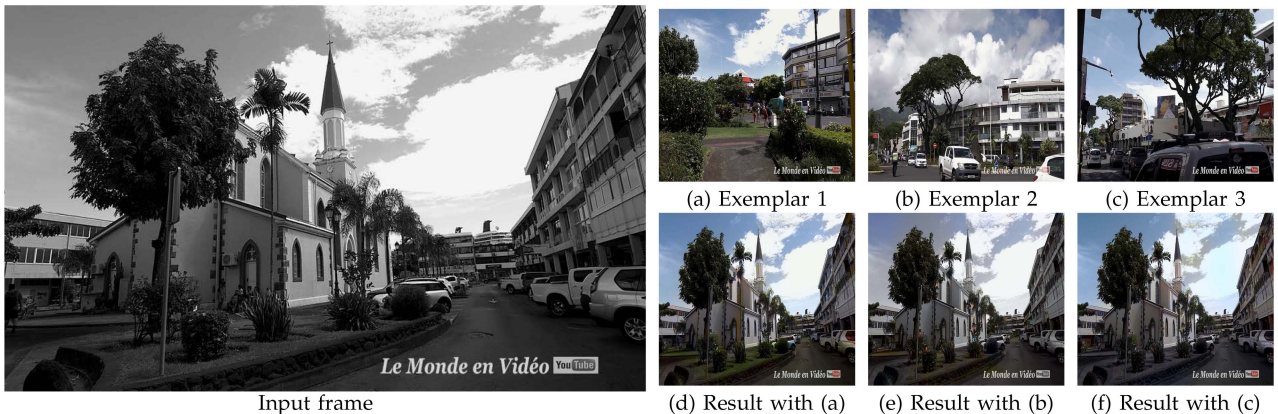


Fig. 14. Qualitative comparisons on the dataset of NTIER 2023 video colorization challenge [1] with different exemplars. The exemplars shown in (a)–(c) are obtained by searching for visually similar images to the input frame on the internet, which have different structures and contents.

the exemplars contain different structures, the BiSTNet is able to generate similar colorization results. This demonstrates that our BiSTNet is robust to the structures of exemplars to some extent.

Exemplars for Real-World Video Colorization Tasks: How to guarantee the quality of exemplars is crucial in real-world scenarios, especially as ground truth exemplars are often not available. One option involves obtaining exemplars through image colorization methods [46], [47], [48]. Fig. 13 shows that our method is able to generate well-colored videos using exemplars generated by various image colorization methods. However, note that these image colorization methods do not consistently guarantee high-quality exemplars. Consequently, the exemplar quality derived from these methods might impact

the performance of our BiSTNet. A more reliable option is to source exemplars from the “Internet” via Google Image Search (<https://images.google.com/>), which offers a rich array of high-quality and vivid images. Fig. 14 demonstrates that even though these exemplars may present diverse structures, our proposed BiSTNet is capable of generating consistent colorization results. This versatility underscores the suitability of our method for real-world applications.

Limitations: The proposed method exhibits satisfactory results in video colorization, yet challenges remain. The original colors of older videos remain unascertained and subjective. Hence, relying solely on the examination of a few exemplars to conduct colorization may not suffice. Efforts to explore domain knowledge of the videos and draw correlations with

TABLE IV
QUANTITATIVE EVALUATIONS OF THE VIDEO COLORIZATION METHODS IN
TERMS OF NETWORK PARAMETERS AND AVERAGE RUNNING TIME

Methods	Parameters (M)	Running time (/ms)
CIC [2]	32.23	339.21
InsColor [12]	103.64	187.40
Gray2ColorNet [44]	53.05	87.41
Color2Embed [43]	176.54	484.44
FAVC [3]	8.73	24.37
TCVC [6]	198.38	61.47
DeepExemplar [5]	59.74	434.78
DeepRemaster [4]	54.28	636.94
BiSTNet (Ours)	158.17	894.85

well-established knowledge pertinent to color schemes of given scenes would yield greater credibility and accuracy in video colorization.

Moreover, although our method achieves favorable performance, it has a higher model complexity. We evaluate our method against other video colorization methods [2], [3], [4], [5], [6], [12], [43] in terms of network parameters and average running time. The average running time is obtained using 34 test images with the size of 832×480 pixels on a machine with an NVIDIA RTX A6000 GPU. Table IV shows that our model has more parameters compared to [4], [5] and the average running time is relatively long.

VI. CONCLUSION

We have proposed an effective BiSTNet to better explore and propagate colors from reference exemplars for video colorization. We establish the semantic correspondence between each frame and the reference exemplars in deep feature space and introduce a bidirectional temporal fusion block that is both simple and effective, enabling improved color propagation. We also develop a mixed expert block to guide the colorization by image prior. To better improve the colorization quality, we progressively colorize frames in a coarse-to-fine manner. We have extensively evaluated our method and show that the proposed BiSTNet performs favorably against state-of-the-art ones on benchmark datasets and real-world videos.

REFERENCES

- [1] X. Kang et al., "NTIRE 2023 video colorization challenge," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2023, pp. 1570–1581.
- [2] R. Zhang, P. Isola, and A. A. Efros, "Colorful image colorization," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 649–666.
- [3] C. Lei and Q. Chen, "Fully automatic video colorization with self-regularization and diversity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3748–3756.
- [4] S. Iizuka and E. Simo-Serra, "DeepRemaster: Temporal source-reference attention networks for comprehensive video enhancement," *ACM Trans. Graph.*, vol. 38, no. 6, pp. 1–13, 2019.
- [5] B. Zhang et al., "Deep exemplar-based video colorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8044–8053.
- [6] Y. Liu et al., "Temporally consistent video colorization with deep feature propagation and self-regularization learning," 2021, *arXiv:2110.04562*.
- [7] A. Levin, D. Lischinski, and Y. Weiss, "Colorization using optimization," *ACM Trans. Graph.*, vol. 23, no. 3, pp. 689–694, 2004.
- [8] Z. Cheng, Q. Yang, and B. Sheng, "Deep colorization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 415–423.
- [9] G. Larsson, M. Maire, and G. Shakhnarovich, "Learning representations for automatic colorization," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 577–593.
- [10] S. Iizuka, E. Simo-Serra, and H. Ishikawa, "Let there be color! joint end-to-end learning of global and local image priors for automatic image colorization with simultaneous classification," *ACM Trans. Graph.*, vol. 35, no. 4, pp. 1–11, 2016.
- [11] R. Zhang et al., "Real-time user-guided image colorization with learned deep priors," *ACM Trans. Graph.*, vol. 36, no. 4, pp. 1–11, 2017.
- [12] J.-W. Su, H.-K. Chu, and J.-B. Huang, "Instance-aware image colorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 7965–7974.
- [13] H. Thasatharan, K. Nazeri, and M. Ebrahimi, "Automatic temporally coherent video colorization," in *Proc. 16th Conf. Comput. Robot Vis.*, 2019, pp. 189–194.
- [14] Z. Xu, T. Wang, F. Fang, Y. Sheng, and G. Zhang, "Stylization-based architecture for fast deep exemplar colorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9360–9369.
- [15] Z. Wan, B. Zhang, D. Chen, and J. Liao, "Bringing old films back to life," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 17673–17682.
- [16] Y. Qu, T. Wong, and P. Heng, "Manga colorization," *ACM Trans. Graph.*, vol. 25, no. 3, pp. 1214–1220, 2006.
- [17] Q. Luan, F. Wen, D. Cohen-Or, L. Liang, Y. Xu, and H. Shum, "Natural image colorization," in *Proc. 18th Eurographics Conf. Rendering Techn.*, 2007, pp. 309–320.
- [18] X. Chen, D. Zou, Q. Zhao, and P. Tan, "Manifold preserving edit propagation," *ACM Trans. Graph.*, vol. 31, no. 6, pp. 1–7, 2012.
- [19] L. Yatziv and G. Sapiro, "Fast image and video colorization using chrominance blending," *IEEE Trans. Image Process.*, vol. 15, no. 5, pp. 1120–1129, May 2006.
- [20] P. Sangkloy, J. Lu, C. Fang, F. Yu, and J. Hays, "Scribbler: Controlling deep image synthesis with sketch and color," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 6836–6845.
- [21] M. He, D. Chen, J. Liao, P. V. Sander, and L. Yuan, "Deep exemplar-based colorization," *ACM Trans. Graph.*, vol. 37, no. 4, pp. 1–16, 2018.
- [22] S. Paul, S. Bhattacharya, and S. Gupta, "Spatiotemporal colorization of video using 3D steerable pyramids," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 27, no. 8, pp. 1605–1619, Aug. 2017.
- [23] B. Sheng, H. Sun, M. Magnor, and P. Li, "Video colorization using parallel optimization in feature space," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 24, no. 3, pp. 407–417, Mar. 2014.
- [24] W.-S. Lai, J.-B. Huang, O. Wang, E. Shechtman, E. Yumer, and M.-H. Yang, "Learning blind video temporal consistency," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 179–195.
- [25] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 402–419.
- [26] J. Park, S. Nah, and K. M. Lee, "Recurrence-in-recurrence networks for video deblurring," in *Proc. Brit. Mach. Vis. Conf.*, 2021, p. 20.
- [27] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1026–1034.
- [28] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: The missing ingredient for fast stylization," 2017, *arXiv:1607.08022*.
- [29] A. Z. Karen Simonyan, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [30] F. Zhan et al., "Bi-level feature alignment for versatile image translation and manipulation," 2021, *arXiv:2107.03021*.
- [31] T. Zhou, W. Wang, E. Konukoglu, and L. Van Gool, "Rethinking semantic segmentation: A prototype view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2572–2583.
- [32] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1395–1403.
- [33] J. Pan, H. Bai, and J. Tang, "Cascaded deep video deblurring using temporal sharpness prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 3040–3048.
- [34] S. Nah, T. Hyun Kim, and K. Mu Lee, "Deep multi-scale convolutional neural network for dynamic scene deblurring," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 257–265.
- [35] F. Perazzi, J. Pont-Tuset, B. McWilliams, L. Van Gool, M. Gross, and A. Sorkine-Hornung, "A benchmark dataset and evaluation methodology for video object segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 724–732.
- [36] E. Kim, S. Lee, J. Park, S. Choi, C. Seo, and J. Choo, "Deep edge-aware interactive colorization against color-bleeding effects," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2021, pp. 14667–14676.

- [37] R. Xu, X. Li, B. Zhou, and C. C. Loy, "Deep flow-guided video inpainting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3723–3732.
- [38] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [39] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 586–595.
- [40] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local nash equilibrium," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6629–6640.
- [41] D. Hasler and S. E. Suesstrunk, "Measuring colorfulness in natural images," in *Human Vision and Electronic Imaging VIII*, Bellingham, WA, USA: SPIE, 2003.
- [42] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [43] H. Zhao, W. Wu, Y. Liu, and D. He, "Color2Embed: Fast exemplar-based image colorization using color embeddings," 2021, *arXiv:2106.08017*.
- [44] P. Lu, J. Yu, X. Peng, Z. Zhao, and X. Wang, "Gray2ColorNet: Transfer more colors from reference image," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 3210–3218.
- [45] C. Lei, Y. Xing, and Q. Chen, "Blind video temporal consistency via deep video prior," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, Art. no. 92.
- [46] Deoldify, 2019. [Online]. Available: <https://github.com/jantic/DeOldify>
- [47] X. Kang, T. Yang, W. Ouyang, P. Ren, L. Li, and X. Xie, "DD-Color: Towards photo-realistic image colorization via dual decoders," 2022, *arXiv:2212.11613*.
- [48] M. Xia, W. Hu, T.-T. Wong, and J. Wang, "Disentangled image colorization via global anchors," *ACM Trans. Graph.*, vol. 41, no. 6, pp. 204:1–204:13, 2022.
- [49] K. C. Chan, X. Wang, K. Yu, C. Dong, and C. C. Loy, "BasicVSR: The search for essential components in video super-resolution and beyond," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 4947–4956.
- [50] K. C. Chan, S. Zhou, X. Xu, and C. C. Loy, "BasicVSR: Improving video super-resolution with enhanced propagation and alignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5972–5981.



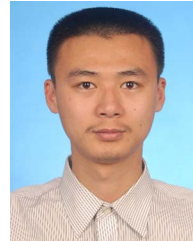
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